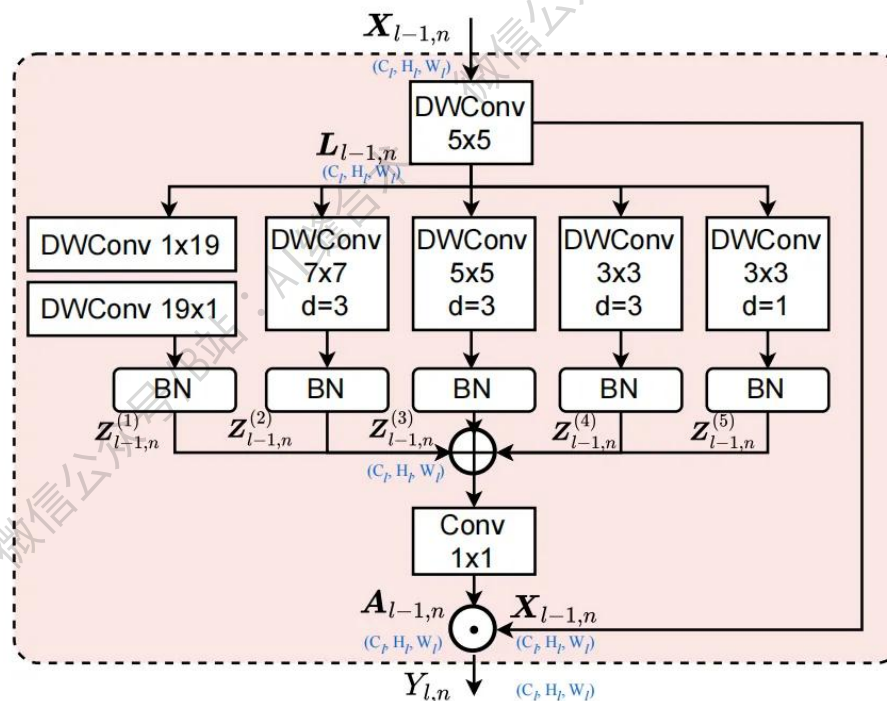


核心速览



(d) PKS Module (§3.3)

论文题目: PKINet-v2: Towards Powerful and Efficient Poly-Kernel Remote Sensing Object Detection

中文题目: PKINet-v2: 迈向强大且高效的多核遥感目标检测系统

论文链接:

<https://arxiv.org/pdf/2603.16341v1>

所属单位: 1 南京理工大学 2 浙江大学 3 成均馆大学电气与计算机工程系 4 先进建筑机械智能制造国家重点

实验室

核心速览:

本文提出 Poly Kernel Inception Network v2 (PKINet-v2) , 一种用于遥感图像目标检测的高效骨干网络。其核心目标是同时解决遥感图像中目标的几何复杂性 (多样宽高比、任意方向) 和空间复杂性 (极端尺度变化、细粒度细节与全局上下文需求) , 并通过创新的重参数化策略提升推理效率。

<https://mp.weixin.qq.com/s/YloyfY7xyt82UDwgfl7f7Q>

```
PKModule(  
  (conv0): Conv2d(64, 64, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=64)  
  (conv1): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1))  
  (branch1_axial): Sequential(  
    (0): Conv2d(64, 64, kernel_size=(1, 19), stride=(1, 1), padding=(0, 9), groups=64, bias=False)  
    (1): Conv2d(64, 64, kernel_size=(19, 1), stride=(1, 1), padding=(9, 0), groups=64, bias=False)  
    (2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  )  
  (branch2_sparse): Sequential(  
    (0): Conv2d(64, 64, kernel_size=(7, 7), stride=(1, 1), padding=(9, 9), dilation=(3, 3), groups=64, bias=False)  
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  )  
  (branch3_sparse): Sequential(  
    (0): Conv2d(64, 64, kernel_size=(5, 5), stride=(1, 1), padding=(6, 6), dilation=(3, 3), groups=64, bias=False)  
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  )  
  (branch4_sparse): Sequential(  
    (0): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(3, 3), dilation=(3, 3), groups=64, bias=False)  
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  )  
  (branch5_dense): Sequential(  
    (0): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=64, bias=False)  
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  )  
)  
input_tensor_shape : torch.Size([2, 64, 128, 128])  
output_tensor_shape : torch.Size([2, 64, 128, 128])
```

哔哩哔哩/微信公众号: AI缝合术, 独家整理!